

AI Workshop 2026

ARTIFICIAL INTELLIGENCE



NSTDA 2026

How LLMs Like ChatGPT Work

Modern AI Foundations



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<https://github.com/kaopanboonyuen/WALDO2026>



 **GitHub**

Two Parts, One Goal: Ready for What's Next

The Modern AI Landscape

- Multimodal AI — what it is
- The LLM concept, in plain terms
- Foundation models — an up-to-date map
- Agentic AI & AI agents
- Tools of the trade: Hugging Face & PyTorch
- Fine-tuning, LoRA, and RAG

From Traditional ML to Modern AI

Weeks 1–7 focused on classical machine learning: engineered features and structured, tabular data. Modern AI shifts to large models trained directly on massive, unstructured data — text, images, and audio.



This week previews Stage 3 and 4 — you will return to full depth after the midterm.

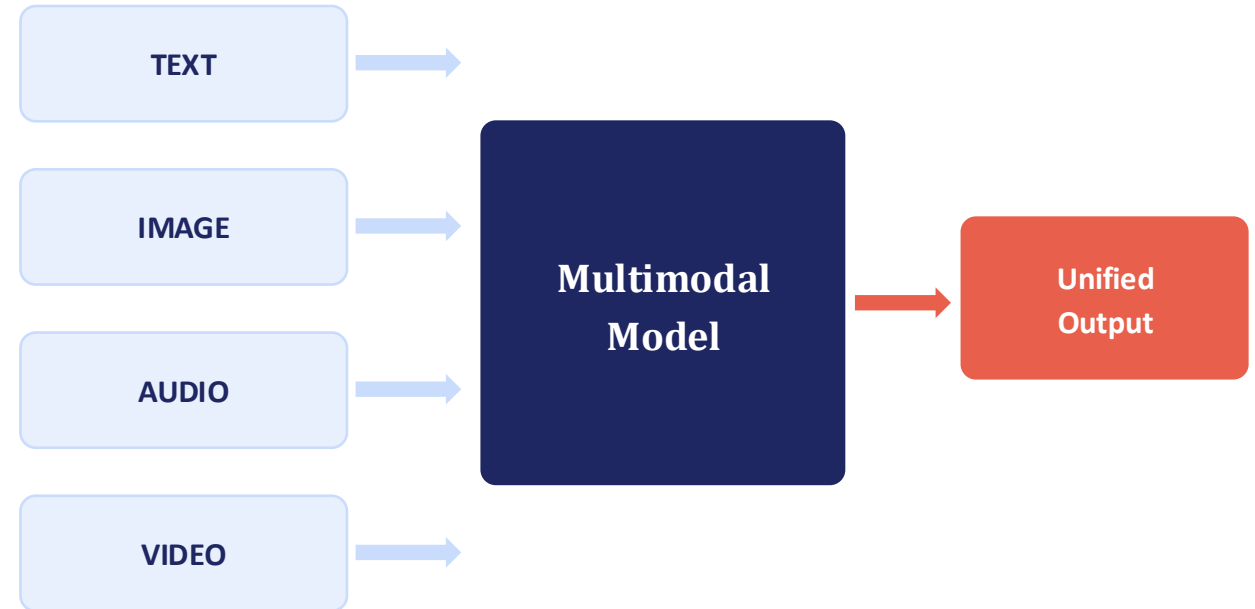
Multimodal AI

Definition

An AI system that can understand and/or generate more than one type of data — text, images, audio, video — and reason across them together, not as separate models bolted together.

Example

Show a model a photo of your fridge and ask, “What can I cook tonight?” — it reads the image and replies in text with a recipe. Or: describe a scene in words and the model generates an image of it.



The LLM Concept

Definition

A Large Language Model (LLM) is a neural network — almost always a Transformer — trained on massive amounts of text to predict the next token (word piece) in a sequence. From this simple task, broad language understanding and generation ability emerges.

Example: “The cat sat on the ____” → the model predicts *“mat”* because that continuation is most probable given everything it learned.

- **Tokens** Chunks of text the model reads/predicts one at a time
- **Parameters** The model's learned weights — billions, for modern LLMs
- **Self-Attention** Lets every token weigh every other token when reasoning

Right: how a base LLM becomes a helpful assistant

Pre-training

Next-token prediction
on huge, general text

Fine-tuning

Trained on curated
instruction examples

Alignment (RLHF)

Tuned to be helpful,
harmless & honest

Deployed Assistant

The chat model
you interact with

Foundation Models — A Snapshot

A foundation model is a single large model pretrained on broad data, then adapted (via prompting, fine-tuning, or RAG) to many downstream tasks. New versions ship constantly — treat this as a snapshot, not a fixed list.

Family	Organization	Known for
GPT (GPT-4 / GPT-4o and later)	OpenAI	General reasoning, multimodal input
Claude (Sonnet / Opus / Haiku)	Anthropic	Long-context reasoning, coding, safety focus
Gemini	Google DeepMind	Native multimodality, very long context
Llama	Meta	Open-weight, widely fine-tuned by others
Mistral	Mistral AI	Efficient, open-weight models

Tip: for the exam, know what a foundation model is and why one model can serve many tasks — you will not be tested on brand names.

Agentic AI: What Is an AI Agent?

Definition: An AI agent is a system built around a foundation model that can **perceive** context, **plan/reason** about what to do, **act** (often by using tools), and **observe** the result — repeating this loop with some autonomy until a goal is reached.

A chatbot answers once. An agent keeps working — in a loop — until the goal is done.

Goal

The instruction or objective it's working toward

Reasoning

The LLM core: plans steps, decides what to do next

Tools

APIs, code execution, web search, files it can call

Memory

Context / history kept across steps of the task

The Agent Loop, in Action



Example: Coding Agent

- Task: “Fix the failing test.”
- Observe: reads the code and the error message
- Think: forms a hypothesis for the fix
- Act: edits the file, re-runs the tests, repeats

Example: Travel-Booking Agent

- Task: “Book me a flight to Chiang Mai on Friday.”
- Observe: checks calendar & budget context
- Act: searches flights via a tool/API, compares options
- Asks for confirmation, then completes the booking

Hugging Face: The Model Hub & Toolkit

Definition

Hugging Face is an open platform and Python ecosystem for sharing and running machine learning models, datasets, and demos. It has become the default place to find and use pretrained models — especially for NLP and, increasingly, all modalities.

- **Hub** Repository of 100,000s of models & datasets
- **transformers** Python library with one common API to load & run models
- **datasets** Library for loading & processing training/eval data
- **Spaces** Hosted demo apps built with a model, shareable via a link

Sample: sentiment analysis in 3 lines

```
from transformers import pipeline

clf = pipeline("sentiment-analysis")
clf("This lecture is great!")

# -> [{'label': 'POSITIVE',
#       'score': 0.999}]
```

PyTorch: The Framework Underneath

Definition

PyTorch is an open-source deep learning framework (originally from Meta AI) built around tensors and automatic differentiation. It is the dominant framework used to build and train most modern foundation models, including the ones behind Hugging Face.

- **Tensor** A multi-dimensional array (like NumPy) that can run on a GPU
- **nn.Module** The building block used to define layers & models
- **Autograd** Automatically computes gradients needed for training

Sample: a tensor and a tiny linear layer

```
import torch
import torch.nn as nn

x = torch.tensor([1.0, 2.0, 3.0])
layer = nn.Linear(3, 1)
y = layer(x)           # forward pass
y.backward()           # autograd!
```

Fine-Tuning a Foundation Model

Definition: Taking a pretrained foundation model and training it further on a smaller, task- or domain-specific dataset, so it specializes for that use case — instead of training a new model from scratch.

Full Fine-Tuning

- Updates all of the model's weights
- Needs the most compute, memory & data
- Produces a full copy of the model per task

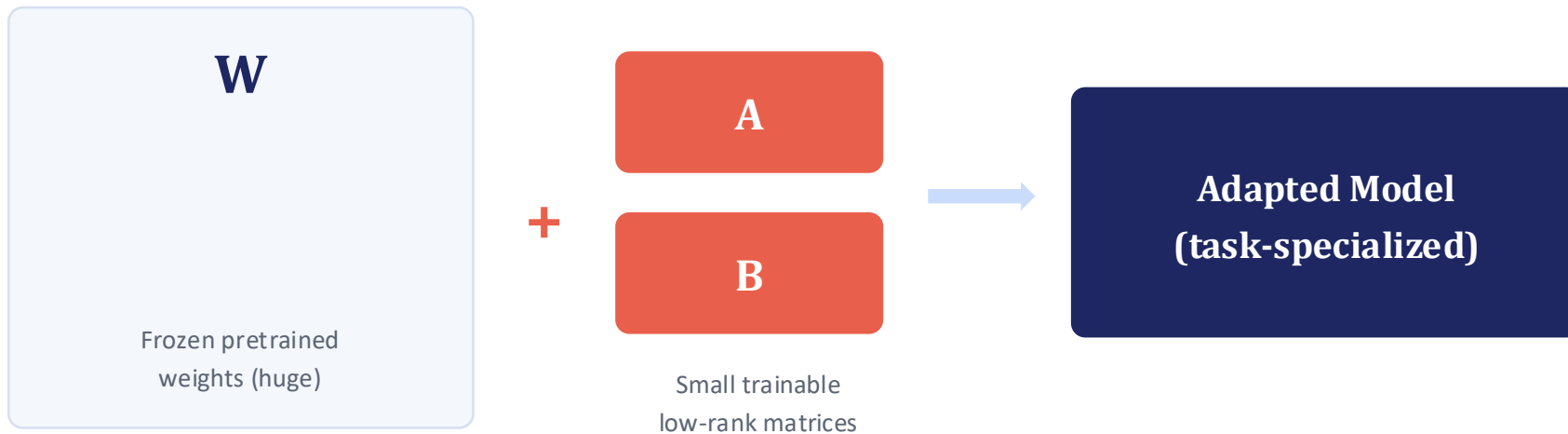
Parameter-Efficient Fine-Tuning (PEFT)

- Freezes most weights, trains only a small added part
- Far cheaper & faster — e.g. LoRA (next slide)
- Small, swappable adapters instead of a full new model

Example: *fine-tuning a general LLM on a Thai bank's customer-service transcripts so it answers in the company's tone and follows its actual policies.*

LoRA — Low-Rank Adaptation

Definition: A PEFT technique that freezes the original weight matrix and injects a pair of small, trainable “low-rank” matrices next to it. Only those small matrices are trained — cutting the compute and memory needed to fine-tune by orders of magnitude.



- Much cheaper & faster than full fine-tuning
- Different LoRA adapters can be swapped on top of the same frozen base model for different tasks
- Often matches full fine-tuning performance for many tasks

RAG — Retrieval-Augmented Generation

Definition: A technique that pairs an LLM with a retrieval step: at query time, the system searches a knowledge base for relevant text, then feeds those passages into the prompt before the model answers — grounding the response in real, current facts instead of only the model's trained memory.



Example: an internal HR chatbot that retrieves the exact policy PDF section before answering an employee's question — instead of the LLM guessing from general training data.

Nine New Terms in One Table

Term	In one line
Multimodal AI	One model that understands/generates several data types together
LLM	Transformer trained to predict the next token; broad language ability emerges
Foundation Model	A large pretrained model reused across many downstream tasks
Agentic AI / AI Agent	A system that observes, plans, acts via tools, and repeats toward a goal
Hugging Face	Hub + libraries for finding and running pretrained models & datasets
PyTorch	The deep learning framework (tensors + autograd) most models are built on
Fine-tuning	Further training a pretrained model on a smaller, specific dataset
LoRA	A cheap fine-tuning method: train small added matrices, freeze the rest
RAG	Retrieve relevant text first, then let the LLM answer using that context



You've built the full picture — classical ML through modern AI.

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